

Problem

The two-wattmeter method gives $P_1 = 1200 \text{ W}$ and $P_2 = -400 \text{ W}$ for a three-phase motor running on a 240-V line. Assume that the motor load is wye-connected and that it draws a line current of 6 A. Calculate the pf of the motor and its phase impedance.

Step-by-step solution

Step 1 of 5

Calculate the total average power, P_T .

$$P_T = P_1 + P_2$$

Substitute 1200 W for P_1 and -400 W for P_2 .

$$\begin{aligned} P_T &= 1200 \text{ W} - 400 \text{ W} \\ &= 800 \text{ W} \end{aligned}$$

Calculate the total reactive power, Q_T .

$$Q_T = \sqrt{3}(P_2 - P_1)$$

Substitute 1200 W for P_1 and -400 W for P_2 .

$$\begin{aligned} Q_T &= \sqrt{3}(-400 \text{ W} - 1200 \text{ W}) \\ &= -1600\sqrt{3} \text{ VAR} \end{aligned}$$

Step 2 of 5

Calculate the power factor angle.

$$\tan \theta = \frac{Q_T}{P_T}$$

$$\theta = \tan^{-1} \left(\frac{Q_T}{P_T} \right)$$

Substitute $-1600\sqrt{3} \text{ VAR}$ for Q_T and 800 W for P_T .

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{-1600\sqrt{3}}{800} \right) \\ &= \tan^{-1}(-2\sqrt{3}) \\ &= -73.897^\circ \end{aligned}$$

Step 3 of 5

Calculate the power factor of the motor.

$$pf = \cos \theta$$

Substitute -73.897° for θ .

$$\begin{aligned} pf &= \cos(-73.897^\circ) \\ &= 0.277 \end{aligned}$$

Thus, the power factor of the motor is 0.277.

Step 4 of 5

Calculate the phase voltage of the wye connected motor.

$$V_p = \frac{V_L}{\sqrt{3}}$$

Substitute 240 V for V_L .

$$\begin{aligned} V_p &= \frac{240}{\sqrt{3}} \\ &= 138.564 \text{ V} \end{aligned}$$

Calculate the phase current of the wye connected motor.

$$I_p = I_L$$

Substitute 6 A for I_L .

$$I_p = 6 \text{ A}$$

Step 5 of 5

Calculate the value of magnitude of phase impedance, Z_p .

$$Z_p = \frac{V_p}{I_p}$$

Substitute 138.564 V for V_p and 6 A for I_p .

$$\begin{aligned} Z_p &= \frac{138.564 \text{ V}}{6 \text{ A}} \\ &= 23.094 \Omega \end{aligned}$$

Calculate the phase impedance of the motor, $\mathbf{Z}_p$.

$$\mathbf{Z}_p = Z_p \angle \theta$$

Substitute 23.094Ω for Z_p and -73.897° for θ .

$$\mathbf{Z}_p = 23.094 \angle -73.897^\circ \Omega$$

Thus, the phase impedance of the motor is $\boxed{23.094 \angle -73.897^\circ \Omega}$.